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Forest carbon sink in the U.S. (1870–2012) driven by substitution of forest ecosystem service flows

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ABSTRACT

Understanding the dynamics behind forest transitions, i.e., shifts from deforestation to forest recovery, is crucial for forest conservation and climate-change mitigation i.e., carbon (C) sequestration. We investigated the drivers of the forest transition in the United States, which was characterized by forest thickening despite surges in industrial wood extraction. We employ the concepts of Human Appropriation of Net Primary Productivity (HANPP) and Material and Energy Flow Analysis (MEFA) to quantitatively assess changes in major provisioning ecosystem services demanded from forests, i.e., industrial wood (comprising biomass used in products such as paper and pulp), grazing, and fuelwood, and analyse substitution processes from 1870–2012 at regional, sectoral, and national scales. The share of industrial wood in total annual forest biomass harvest increased from 23% to 84% over the time-period, while fuelwood and biomass grazed declined from 63% to 13%, and 14% to 3%, respectively. Reductions in demand for fuelwood and biomass grazed were enabled by shifts in feed and energy sources, consequently allowing for increases in both livestock numbers and energy use. Feed crops increased six-fold, alleviating grazing pressure on forest ecosystems, particularly in the Eastern states. Fossil fuels replaced fuelwood, especially in the residential sector. Between 1900–2012 the final energy mix increased seventeen-fold. Thus, the increase in biomass C stocks in U.S. forests was connected to substitution of forest ecosystem services with fossil fuel-based production systems, and with manifold increases in societal resource use and C dynamics. Such shifts need to be considered when assessing the positive environmental effects of forest transitions.

1. Introduction

Conserving forests is a major aim in climate-change mitigation and biodiversity conservation. In this context, it is important to understand the drivers of forest transitions (FTs), i.e., the long-term shift from large-scale deforestation to reforestation (Kauppi et al., 2006; Mather, 1992; Mather and Needle, 1998). Reforestation can either occur through natural succession, e.g., when agricultural land is abandoned, or through tree planting (Lambin and Meyfroidt, 2010), which, if established as monocultures, may have negative implications for forest health, biodiversity, and ecosystem resilience (Biedermann et al., 2019; Heilmayr et al., 2016; Yu et al., 2019). FTs have been identified in many countries around the world, particularly in the context of industrialisation (e.g. Gingrich et al., 2016; Kuemmerle et al., 2015; Le Noë et al., 2020; Li et al., 2017; Singh et al., 2017). Conceptual work has focused on

identifying specific “forest transition pathways” (Lambin and Meyfroidt, 2010) based on land-use, socio-economic or political enabling processes (Barbier et al., 2010; Meyfroidt and Lambin, 2011; Perz, 2007; Rudel et al., 2005). By contrast, FTs have only recently been studied from a socio-ecological perspective, with a focus on related changes in societal resource use (Gingrich et al., 2019, 2021; Le Noë et al., 2020; Pichler et al., 2020), and have rarely been connected to changes in ecosystem services (Satake and Rudel, 2007; Wilson et al., 2017). Societal use of forests can alter significantly due to changes in demand and supply of forestry products and has an impact on the carbon balance of wood products (Sathre and Gustavsson, 2006). Understanding these interconnections is important for designing effective forest conservation and management policies that consider trade-offs between forest functions (Lin and Ge, 2020) and avoid problem shifts to other sectors (Gingrich et al., 2019; Scheidel and Gingrich, 2020).

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Given the rich historiography on forests and broad historical data availability, the United States offer a unique opportunity to study interconnections between long-term forest change and resource use. There is a long history of land-use in North America, where forests have been used long before European Colonisation in the 1500s (Pyne, 1997). The (USDA Forest Service 2000a) estimated that since 1630, around one third of the original forest area has been cleared, mainly due to land-use conversion to croplands, permanent pastures, or built-up infrastructure. Additionally, some of the remaining old-growth forests were replaced by intensively managed tree plantations, especially in the South-East (Grelen, 1978; Oswalt et al., 2014). The timing and location of these changes varied considerably across the U.S. and some occurred well before consistent documentation was available (Birdsey et al., 2006; Frederick and Sedjo, 2013; Gregg, 2010; Ramankutty et al., 2010; Waisanen and Bliss, 2002). From the late 19th to the early 20th century, forests were vitally important for providing resources to the industrialising economy, namely construction materials (timber for e.g. houses or fences, MacCleery, 1993; Wernick et al., 1997) and energy (fuelwood for e.g. heating and cooking, O'Connor and Cleveland, 2014; Richter et al., 2009; Williams, 1987, 1992). In addition, forests fulfilled other functions essential for rural livelihoods, such as providing feed for livestock (forage, understorey) (Fedkiw, 1989; Grelen, 1978; USDA Forest Service, 1956). However, extensive “clearcutting” led to severe degradation of forest ecosystems (Adams, 1975; Galbraith and Anderson, 1971; Haugo et al., 2010; Miller and Halpern, 1998), especially in the Eastern section of the U.S. (Houghton, 1999; Williams, 1987), and unregulated forest grazing led to conflicts over public land (Borman and Johnson, 1990; Kosco and Bartolome, 1981; Prevedel and Johnson, 2005). Provisioning functions of forests were strongly altered since the beginning of the 20th century, as new forest policies and management practices were established (Pinchot, 1905; Steen, 2004; USDA Forest Service, 1948). In this period, the U.S. transitioned from a mainly biomass-based to an industrial energy system, emerging as one of the largest consumers of non-renewable resources, in particular fossil fuels, on the planet (Gierlinger and Krausmann, 2012; Krausmann and Fischer-Kowalski, 2010). Although forests continue to be important for the economic situation of numerous people in the U.S. (Egilmmez et al., 2014), the way forests are used has fundamentally changed (Foell, 2019; Warde, 2019; Williams, 1987), consequently stabilising and increasing forest biomass stocks for most of the 20th and early 21st centuries (Birdsey et al., 2006; Magerl et al., 2019). However, the underlying biophysical processes connected to these historical developments in forest change have, to our knowledge, never been quantified and remain poorly understood.

In this study, we use the Ecosystem Services (ES) framework to comprehensively address the changing socio-ecological interconnections of forest use and shifts in societal resource use (Deng et al., 2013). ES include provisioning (e.g., feed, wood, clean water, fibres), supporting (e.g., photosynthesis, nutrient cycling, soil formation), regulating (e.g., carbon (C) sequestration, protection against floods), and cultural services (e.g., recreation, spiritual, aesthetic), denoting the actual (non-)physical benefits people draw from ecosystems (Millennium Ecosystem Assessment (Program), 2005). We assess changes in provisioning ES, i.e., different types of biomass products derived from forests, and link them to a) changes in forest ecosystem service capacity (forest area, biomass C stocks, and NPP), and b) ecosystem service demand, i.e., processes outside the forestry sector influencing the developments of these ES in particular. In the U.S., like in many countries, the provisioning ES by forests shifted from multiple uses by mostly rural populations to industrial wood provision in the course of the FT (Pichler et al., 2020; Prudham, 1998). In parallel, the U.S. have implemented forest conservation policies, enhancing ES capacity by increasing C sequestration in forests (Alkama and Cescatti, 2016; Duveiller et al., 2018; Keenan and Williams, 2018). To study the changes in provisioning ES and their relevance for forest dynamics, it is important to address temporal shifts in ES and their underlying drivers (Rau et al., 2020; Schirpke et al., 2019). Here, we contribute to this systemic

understanding by assessing shifts in forest ES capacity and ES demand from a long-term perspective, covering the entire period for which information on forest area and timber volumes is provided by contemporary inventories. Precisely, using socio-ecological indicator frameworks (see Section 2.2), we investigate changes in provisioning ES over the course of the FT in the U.S., covering the period 1870–2012, and link them to shifts in ES demand associated to resource use outside of the forest sector i.e., shifts in feed and fuel use. Our study thus uncovers the links between long-term forest change and resource use, contributing to a better understanding of sustainability implications of FTs.

2. Concepts, methods and data

2.1. Ecosystem services: capacity, demand, and substitution processes

Conceptually, our study builds on the framework by Tomscha et al. (2016), which distinguishes between ES capacities and ES demand, connected by ES flows and ES feedbacks (Fig. 1).

We used official reports and statistics (Defebaugh, 1907; Eggleston, 1885; Fedkiw, 1989; Greeley, 1920; USDA Forest Service, 1932, 1948, 1956; Smith and Pierson, 1918), as well as historic and contemporary qualitative sources on forest use (Adams, 1975; Clawson, 1979; Corder, 1973; Gregg, 2010; Grelen, 1978; Kosco and Bartolome, 1981; MacCleery, 1993; Prevedel and Johnson, 2005; Steen, 2004; Williams, 1987, 1992) to identify and quantify distinct patterns of forest use, as well as their diverging impacts on forest biomass on the national, and where possible, on the regional scale.

We scrutinize flows of three main provisioning ES of U.S. forests by quantifying the amount of biomass extracted annually through three major practices: (1) *Forest grazing* i.e., biomass consumption by grazing livestock; (2) *Industrial wood harvest*, i.e., for products such as paper and pulp, lumber, veneer, etc.; (3) *Fuelwood* extraction for heating, cooking, and mechanical drive (steam trains and -boats). ES capacity (Fig. 1) in our study is translated into the underlying variables enabling the provisioning ES flows i.e., forest area, biomass net primary production (NPP) and biomass C stocks of forests (excluding C stock in soils). ES demand (in biophysical, not monetary terms) refers to the three major sectors demanding provisioning ES flows from forests: the livestock system, the timber industries, and final energy consumers. Observed changes in forest biomass and productivity can be interpreted as ES feedbacks caused by the demand for ES provisioning flows. Beyond implicitly addressing one regulating forest ES, i.e. the annual C sequestration in forest biomass as the difference in C stocks between two points in time (Eggleston, 2006), we do not cover any further regulating or cultural forest ES, or other feedbacks e.g., biodiversity loss/gain, soil compaction.

Instead, we investigate shifts in external drivers that occur outside of but in relation to forests: We analyse changes in livestock feed in quantitative and qualitative terms, i.e., the respective shares of forest and grassland biomass, feed crops and residues in total feed on sub-national scales. Similarly, we assess changes in the final energy consumption in households and economic sectors. We ask how shifts in demand for provisioning ES have influenced the actual flow of biomass in forests and how they relate to changes in ES capacity, i.e., forest area, NPP and biomass C stocks.

Foreign trade of wood can be a major factor impacting carbon flows (Ståhls et al., 2011), the relationship of ES demand and supply, as well as substitution processes (Gingrich et al., 2019). However, we argue that in the U.S., changes in ES demand may be explained by inherent shifts in domestic consumption because trade did not play a significant role in reducing pressures on forests during the observed period: Imports of feedstuff represented only a small amount (~1%) of total livestock feed (Gierlinger and Krausmann, 2012; USDA National Agricultural Statistics Service, 2020). Industrial wood imports amounted to an average of 8% of the total domestic wood consumption (production minus exports plus

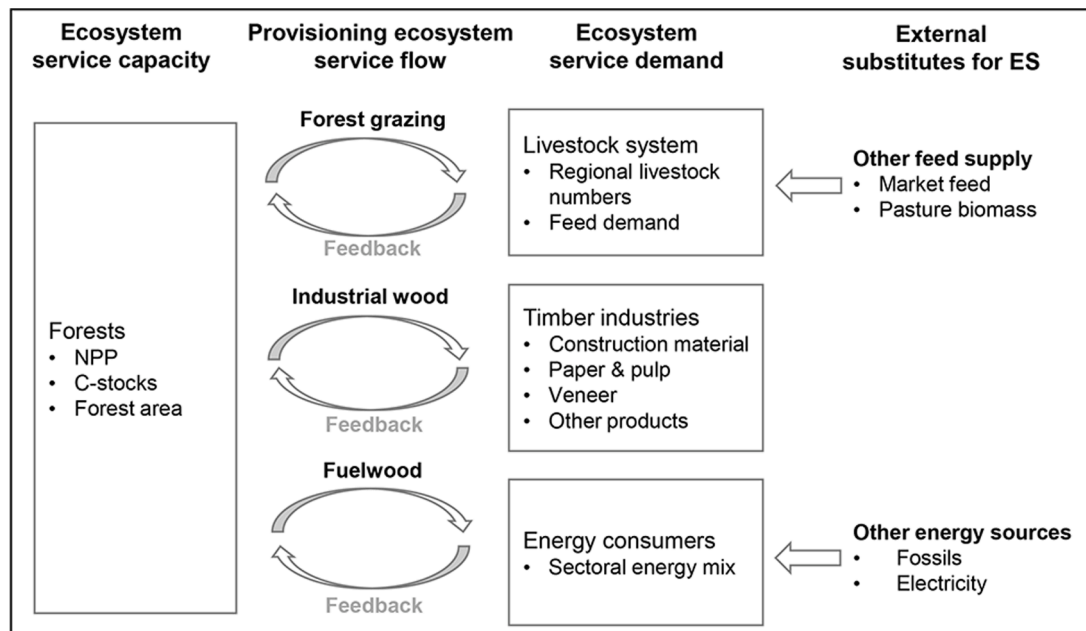


Fig. 1. Components of the reconstruction of forest ecosystem services dynamics in the U.S. 1870-2020 in this study. Conceptual scheme adapted from original figure by Tomscha et al., (2016, S. 749 Fig. 1); Own figure.

imports) between 1900 and 1990. Only in 2000 and 2007, wood imports peaked at 17% and 18% respectively (Department of Commerce and Labor, 1908; FAO, 2020; Howard, 2007). Therefore, we focus on domestic processes only, and neglect the role of foreign trade.

2.2. Methods and data

We consistently investigate forest ES capacity, flows, and demand for the conterminous U.S. (48 states) in the period 1870-2012. ES capacity (forest area and biomass stocks) from 1910-2012 reported here stems from a previous study based on U.S. timber inventories (Magerl et al., 2019), while trajectories of forest net primary production (NPP) for the same period were derived from a dynamic vegetation model (LPJ-GUESS, Lund-Potsdam-Jena General Ecosystem Simulator) (Smith et al., 2014). Although several studies reconstructed forest composition on regional scales in the U.S. before 1910 (e.g., Jeon et al., 2014; Rhemtulla et al., 2007), to our knowledge, no consistent databases exist that would allow us to infer the national trend in biomass C stocks prior to 1910, hence limiting the results presented here to 1910 onwards. To assess the three major provisioning ES flows, we consistently quantify the amount of above- and belowground forest biomass extracted for grazing, fuelwood, and industrial wood use (see also 2.2.1.). By using national statistics and country-specific expansion factors to quantify annual biomass extraction, we implement a consistent IPCC tier 2 approach (Eggleston, 2006), although belowground- are inherently more uncertain than aboveground compartment estimates, as even field studies hardly measure this forest compartment (Schepaschenko et al., 2017). In addition, we analyse shifts in societal resource use enabling declining ES demand for feed (at the sub-national level) and fuelwood (at the sectoral level), in parallel to a surge in industrial wood demand and forest biomass stocks. We here describe the principal data sources and estimation procedures. For a more detailed description of data and methods, see supplementary information (SI).

2.2.1. Changes in provisioning forest ecosystem services

In order to assess the provisioning ES flows of U.S. forests, we apply the indicator framework “Human Appropriation of Net Primary Production” (HANPP) (Haberl et al., 2007; Kastner, 2009; Krausmann et al., 2013). HANPP quantifies the combined impact of land conversion

($HANPP_{luc}$) and biomass extraction ($HANPP_{harv}$) on the NPP of ecosystems and can be expressed in tons carbon per year (tC/yr) or % of the potential net primary production (NPP_{pot}) i.e., the productivity of the ecosystem that would prevail hypothetically in the absence of land use.

We here quantify total extraction of forest biomass ($HANPP_{harv}$) induced by forest grazing, fuelwood extraction and industrial wood harvest. We compare the respective levels and quantitative impacts of each provisioning ES flow on forest NPP at 17 points in time in the studied period. Supplementary table S1 presents the HANPP fluxes corresponding to the three respective provisioning ES flows analysed in this study. The sub-national disaggregation of our analysis of trends in forest grazing is based on information of the United States Forest Service (USFS) (Table S2, Figure S1).

While the quantification of forest grazing relies on an assessment of feed demand and livestock numbers, fuelwood and industrial wood harvest could be inferred directly from forest inventories and extraction statistics but only at the national level, since no comprehensive database is available for the entire timeframe on sub-national scales (FAO, 2020; Manthy and Potter, 1978; United States Bureau of the Census, 1975). Because these data refer to final wood i.e., merchantable roundwood (FAO, 2020), we applied expansion factors, accounting for losses such as branches, bark, slash, and belowground biomass (roots) destroyed during harvest (Krausmann et al., 2013). All values were converted to tC/yr, using an average C content of 0.5 for dry matter of biomass (Gingrich et al., 2016; Schandl et al., 2002).

We derived region- and land-cover specific per area values for NPP_{act} (the NPP of the actual vegetation) and NPP_{pot} from the LPJ-GUESS model (sections S3, S4.3.1, Smith et al., 2014), taking into account temporal dynamics of NPP caused by climate change and CO_2 fertilization. Based on Klein Goldewijk et al. (2016), grazing land was split into “natural” grassland i.e., rangelands, and “artificial” grassland i.e., pastures. Forests were differentiated into “open” forests (canopy cover 15-40%) and “closed” forests (canopy cover >40%) based on ESA CCI land cover (CCI, 2017), for each of which the NPP_{pot} and NPP_{act} values of LPJ-GUESS varied across the United States. Following commonly-employed rationales (Erb et al., 2016; Noormets et al., 2015), we assume $HANPP_{luc}$ on forests to equal zero i.e., NPP_{pot} equals NPP_{act} (Haberl et al., 2007; Krausmann et al., 2013).

2.2.2. Changes in the livestock system

Changes in the livestock system were analysed to understand shifts in feed supply and arrive at a robust estimate for forest grazing, based on a “grazing gap” approach (Krausmann et al., 2013). To this end, we compiled numbers of different livestock species on the state level and converted them to “livestock units” (LU), using published factors (Table S3). We separated livestock into grazers (cattle, horses, mules, asses, goats, and sheep) and non-grazers (poultry, pigs). Next, we estimated the total feed demand per year and species by multiplying LUs (except for cattle, poultry and pigs) by daily species-specific feed intake factors (Table S4). For the feed demand of cattle, poultry, and pigs, we used time-specific factors, based on animal product statistics from the USDA National Agricultural Statistics Service (2020) and feed conversion efficiencies i.e., feed intake per final product (Table S5) under consideration of feed efficiency trajectories (Fricko et al., 2017, Table S6). This way, we accounted for “efficiency improvements” due to breeding and genetic engineering in intensive livestock production systems. Next, we quantified the region-specific feed supply per year, adding data for feed crops, hay and straw, and other feedstuff (e.g., fish, demersal) reported in USDA data sources (section S4). We first attributed the feed supply to non-grazers and then to grazing animals. The difference between reported feed supply and feed demand of grazing animals as specified above, i.e., the missing amount of biomass in order to meet their total feed demand, was assumed to be grazed directly on pastures and in forests (Krausmann et al., 2013).

We compiled data on forest area grazed, reported at the national scale from 1880–2012 and on the state level from 1945 to 2012 (Bigelow and Allison, 2017). Estimation of 1870 grazed forest area was based on qualitative information. Areas of pastures 1945–2012 were derived from Bigelow and Allison (2017). For 1870–1945 we used a similar approach as for grazed forest area. Area-specific values of NPP_{pot} and NPP_{act} for the respective land uses (forest area grazed, natural and artificial pastures) were derived from the LPJ-GUESS on the national scale for 1890–2012 and on the sub-national scale for 1910, 1930–2012 (Smith et al., 2014), and multiplied by the area shares of pastures and forests to calculate the respective total annual NPP_{pot} and NPP_{act} . Missing years were linearly interpolated. We validated our estimate by comparison to the HYDE land-use database from 1910 to 2012 (Klein Goldewijk et al., 2016) and additional sources for 1870–1910 (Olson et al., 2001; Ramankutty and Foley, 1999; Section S3).

To differentiate the amount of biomass grazed in forests vs. pastures on regional scales, we split total biomass grazed into respective shares for each ecosystem. Since neither robust spatially explicit data on biomass grazed by livestock nor stocking numbers in forests or pastures were available for the whole timeframe studied, we estimated three different scenarios (SI 4.3.) representing a possible range for livestock grazing in forests. For each scenario we made diverging assumptions on forest grazing intensities, defined as grazing pressure as a fraction of forest NPP_{act} (Fetzel et al., 2017). In scenario 1 (section S4.3.2.) we assumed biomass grazed per area on forests and pastures to be equal by dividing the total biomass grazed by total area grazed and multiplying it by the respective share of each ecosystem. For scenario 2 (section S4.3.3.) we divided total grazed biomass by total NPP_{act} of grazed forests and pastures, and multiplied by the respective NPP_{act} of each ecosystem, representing the upper boundary of our estimate. Scenario 3 is based on temporal and spatially limited statistics, thus representing the lower boundary of our estimation: Grazing statistics report the numbers of grazers in national forests in 37 states between 1966–2012 (Range Management, 2013; USDA Forest Service, 1969, 1971, 1981, 1990, 2000b, 2008). For the missing years 1870–1966, we linearly extrapolated values from 1966 backwards (section S4.3.4.). The derived livestock numbers were used to estimate biomass grazed, as described above.

2.2.3. Changes in the energy system

To contextualise the reductions in fuelwood extraction and

associated substitution thereof with fossil fuels, we conducted a Material and Energy Flow Analysis (MEFA) (Fischer-Kowalski et al., 2011), quantifying final energy consumption of fuelwood and other sources of energy (coal, oil, natural gas, electricity) over time on the national scale, disaggregated into the four main energy-end-use sectors: The industrial sector, the transport sector, the residential, and the commercial and public sector for which data between 1950–2012 were readily available (International Energy Agency, 2019; U.S. Energy Information Administration, 2020, 2011). We reconstructed sectoral fuel use before 1950 using information from the United States Bureau of the Census (1975) and other studies (Fishlow, 1966; Gschwandtner et al., 1986; Manthly and Potter, 1978; Morrison, 1992; U.S. Congress, Office of Technology Assessment, 1983; Warr et al., 2010; Williams, 1987). We did not consider the energy transformation sector and thus, fuelwood used for energy transformation purposes (e.g., electricity generation) because according to Corder (1973) and Schurr (1984), before 1950 wood was almost exclusively used for household- (and to a much smaller extent for industrial) heating purposes (section S6).

2.2.4. Sensitivity analysis

To assess the robustness of our results we conducted a sensitivity analysis (section S7). By applying published negative and positive deviation values, we tested the main input factors possibly distorting our results. We derived an uncertainty range for the respective HANPP flows assessed in this study, representing plausible variance related to deviations in statistical sources and underlying methodological assumptions (Krausmann et al., 2013). Additionally, we evaluated the effect enhanced plant productivity due to climate change and CO_2 fertilisation had on ecosystem productivity: We calculated an alternative HANPP estimate under the assumption that NPP_{pot} would have remained unchanged throughout the observed period.

3. Results

The forest transition in the U.S. was characterised by major shifts in ES capacity and demand since 1870. Fig. 2a shows the change in ES capacity - total forest biomass C stocks, forest area (Magerl et al., 2019), and forest $NPP_{pot}/area$ (Smith et al., 2014) - in the U.S. relative to 1910. Biomass C stocks almost doubled between 1910 and 2012 (+80%), increasing from 10.2 PgC (Petagram C) to 18.4 PgC, while total forest area on the national scale remained relatively unchanged between 1910 and 2012, indicating forest thickening i.e., increase of forest biomass density, in tC/ha (hectare). Forest area had been shrinking from 303.0 mio (million) ha in 1870 to around 257.5 mio ha in 1910 caused by land-use change, including agricultural westward expansion, and partly abandonment of cropland in the Eastern U.S. (Cunfer, 2005; Foster et al., 1998; Ramankutty et al., 2010). According to the LPJ-GUESS dynamic vegetation model (Smith et al., 2014), $NPP_{pot}/area$ increased by almost 20% in the period observed, caused by environmental change. Increase of NPP_{pot} has been reported across dynamic vegetation models (Pachauri et al., 2014; Walker et al., 2021; Zhu et al., 2016). Although enhanced NPP_{pot} might influence actual growth of forest biomass (i.e., the difference between gross annual increment, natural mortality, harvest and disturbances such as fire) to some extent due to altered growth conditions, this development is not directly translatable into changes of observed forest biomass growth (Hickler et al., 2015; Wang et al., 2020). The total demand for all provisioning ES flows decreased from 1910–1970 by almost 40%, and reached around 90% of its 1910 value again in 1990 (Fig. 2a). The coinciding developments of increasing $NPP_{pot}/area$ and declining total ES demand contribute to an increasing ES capacity and signify a slight relief from pressure on forests to deliver provisioning ES.

Total ES demand (total forest $HANPP_{harv}$), i.e., the sum of biomass extracted annually through forest grazing, fuelwood extraction and industrial wood harvest peaked in 1910 at 200.9–225.0 (scenario 1–3) TgC/yr (Teragram C per year) and again in 1990 at 194.3–198.7 TgC/yr

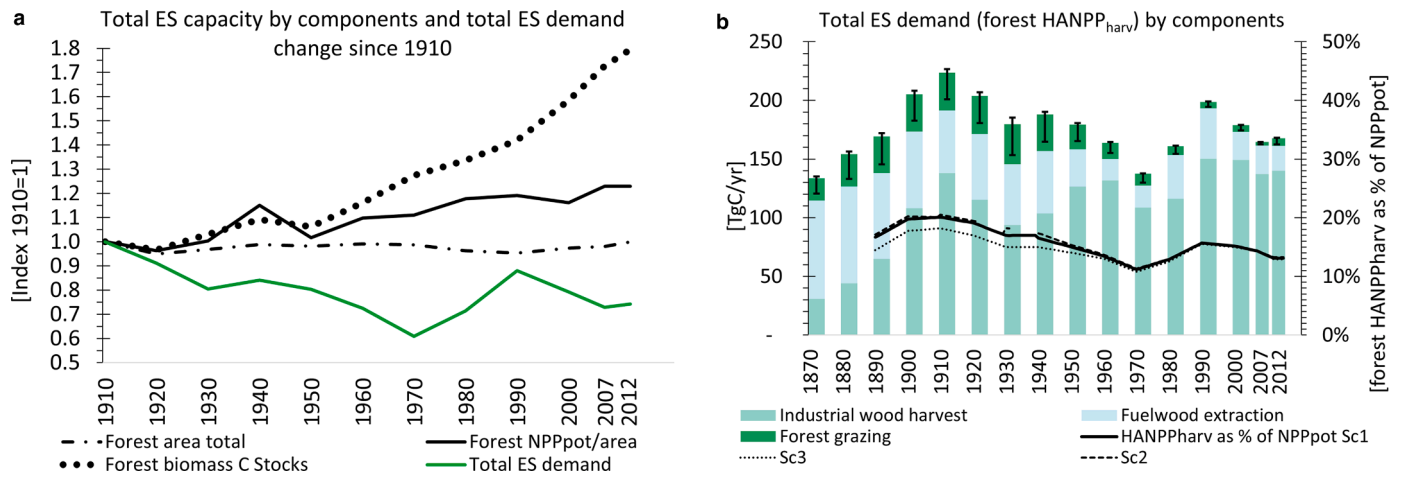


Fig. 2. Forest change in the conterminous U.S. **a.** Total ES capacity 1910-2012 by components (forest area, biomass C stocks, NPP_{pot}/area), and total ES demand, relative change indexed to 1910=1 **b.** Total ES demand (total forest HANPP_{harv}) 1870-2012 as share of NPP_{pot} in forests [solid (forest grazing scenario 1), dashed (scenario 2), and dotted (scenario 3) line, secondary axis, %] and by components (ES flows): fuelwood extraction, industrial wood harvest, and forest grazing in absolute numbers [bars, primary axis, TgC/yr]; Notes: Biomass C stocks available from 1910 onwards (Fig. 2a); NPP_{pot} available from 1890 onwards, error bars indicate minimum and maximum variations for total forest HANPP_{harv}, caused by the three scenarios for forest grazing (Fig. 2b).

(Fig. 2b). Between these two peaks it fluctuated between 129.8–136.8 (scenario 3) and 180.7–204.9 TgC/yr (scenario 2). After the second peak in 1990, total forest HANPP_{harv} decreased to 162.4–179.1 TgC/yr (scenario 1–3). Due to the fluctuating total forest HANPP_{harv}, and simultaneously increasing NPP_{pot}, total forest HANPP_{harv} as a percentage of NPP_{pot} in forests declined from a maximum of 18–20% in 1910 to 10–11% until 1970, increased until 1990 to 15–16% and declined again to 12–13% in 2012 (scenarios 1–3, solid, dashed and dotted black lines in Fig. 2b). The three different assessments of forest grazing cause maximum variations of -84% (scenario 3, 1980) to +64% (scenario 2, 2007) (error bars in Fig. 2b) in forest grazing, leading to maximum fluctuations in total forest HANPP_{harv} of -13% (1890) to +3% (1880).

In the 142 years studied, total forest HANPP_{harv} fluctuated in absolute terms, and the composition of provisioning forest ES flows changed significantly, shifting from mainly energy provision (fuelwood) to mainly material provision (industrial wood) (Fig. 2b). While in 1870, fuelwood extraction accounted for 83.7 TgC/yr, or 61% of total forest HANPP_{harv}, its importance declined significantly until 1960 (18.0 TgC/yr, or 12% of the total). However, the share of fuelwood in total forest HANPP_{harv} grew again after 1970, to 13–19% between 1990-2012. By

contrast, the share of industrial wood harvest in total forest HANPP_{harv} tripled from 1870/1910, to 59% or 138.1 TgC/yr, an increase of 107.2 TgC/yr in just 40 years. Thereafter, its share declined slightly to 48% of total forest HANPP_{harv} in 1930, but increased again afterwards and remained between 50 and 70% for the rest of the observed period. According to our estimations, the share of forest grazing in total forest HANPP_{harv} was lower than the other two ES investigated, contributing to 5–15% of the total between 1870 and 1940. However, after World War II the share of forest grazing declined to 1–3% of total forest HANPP_{harv} in 1990 and remained relatively constant until 2012.

3.1. ES demand for forest grazing

While the area of total forest remained relatively unchanged over the period studied (Fig. 2a), the share of total forest used for grazing declined significantly, from an estimated 173.9 mio ha (more than 57% of the total forest area) in 1870 to 52.5 mio ha in 2012 (20% of the total). This can be observed in all four main geographical regions of the conterminous U.S. analysed in this study (Fig. 3a, Fig. S1, Figs S2a–d), with the strongest reduction in the South (-94%) and the least

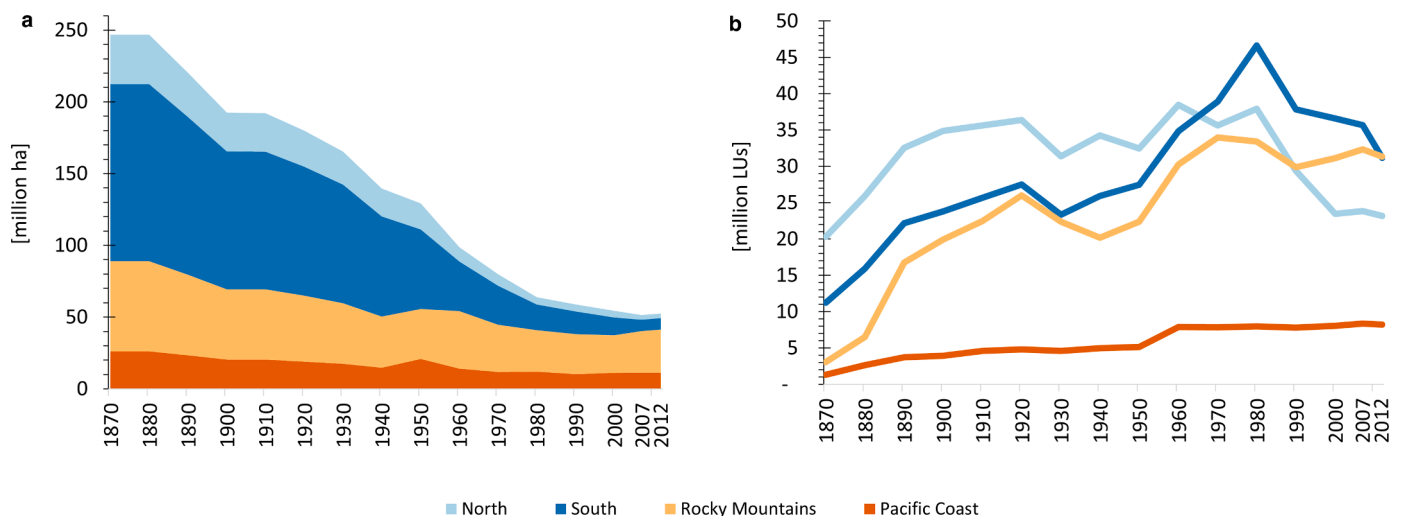


Fig. 3. Regional disaggregation of **a)** total forest area grazed [million hectare] and **b)** numbers of grazers (cattle, goats, sheep, horses) [million livestock units, LUs]; Conterminous U.S., 1870-2012

pronounced but still significant reduction in the Rocky Mountains (-52%), where forest grazing area even increased slightly (+9%) from 2000–2012 (Fig. 3a). At the same time, numbers of grazers increased or remained high through most of the observed time-period in the four regions (Fig. 3b). Cattle accounted for 50–70% of total livestock units (LUs) (Fig. S3) in the U.S. throughout the entire timeframe studied. Between 1870 and 1920, the number of grazers in the country had almost tripled from 35.9 mio LUs to 94.6 mio LUs. After a decline during the great depression of the 1930s, LUs increased again, especially after WWII. A second peak was reached in 1980 (125.8 million LUs), followed by a decline until 1990. LUs remained between 93.8 and 104.8 mio afterwards. Regional disaggregation of LUs (Fig. 3b) reveals that relatively similar trends on different levels occurred in all regions until 1960. However, after LUs peaked between 1960 and 1980 nationwide, in the North and South the number of grazers declined sharply, while in the Rocky Mountains they increased again slightly until 2007. In the Pacific Coast region, LUs stabilised in 1960 and remained unchanged for the rest of the period.

The decline in forest grazing (Fig. 4b–f) and simultaneous increase in LUs (Fig. 3b) were enabled by substantial changes in the composition of total livestock feed, representing the industrialisation of the livestock system (e.g., increased agricultural productivity, establishment of feedlots). The largest shares of total feed were consumed in the North and South (Fig. 4a) throughout the timeframe studied. In terms of feed composition, relative shifts from forest grazing to external substitutes (feed crops plus residues, and hay) occurred in all regions. Figs 4b–f show total biomass consumed by grazers in forests (dark green areas representing scenario 1, and error bars for scenario 2 and 3, respectively), on pastures (light green areas), feed crops and residues (yellow areas), and hay (brown areas). On the national scale (Fig. 4b), forest grazing peaked in 1910 at 9.6–33.1 TgC/yr or 8–23% (Scenarios 3 and 2) of total feed intake and declined to 5.4–5.8 TgC/yr or 3% respectively, in 2012. Forest grazing intensity increased from 1% to 5% of NPP_{act} in forests in 1890 to 1–7% in 1930, and declined to around 1% (all three scenarios) in 2007. These trajectories in national forest grazing intensity can be explained by shifts in total feed supply to feedstuff and pastures, in parallel to the decreasing areas of grazed forests (Fig. 3a, Fig. S2). The slight increase of forest grazing intensity in 2012 to <1%–3% can be explained by the fact that the number of cattle (which consume the largest shares of grazing biomass among grazers) increased or stabilised in three of four regions (Fig. 3b) in parallel to the decrease in forest area grazed.

Fig. 4c–f present total feed for grazers estimated for the four respective regions, revealing distinct regional livestock feeding systems: In the North, Rocky Mountains and Pacific Coast, (Fig. 4c,e,f) forest grazing accounted for respective maximum shares in total feed supply (scenario 3 and 2) of 22–25% in 1880, 9–15% in 1870, and 7–31% in 1870. In these regions hay and pastures dominated feed compositions throughout the entire time-period, and forest grazing was obsolete after 1950. On the contrary, in the South (Fig. 4d) the share of forest grazing in total feed supply was considerably higher (maximum of 44–66% in 1870), never decreasing below 5% throughout the observed period. Despite the different relative compositions of feed supply in the four regions, the shares of feed crops and hay increased significantly after WWII in all regions except for the South: Here, pastures and hay supplied most of total feed after 1880, and the share of crops was relatively low (a maximum of around 12% in 1980). While regional trajectories of forest grazing intensities in the Rocky Mountains and Pacific Coast resembled the national trend, albeit at lower levels, there were significant differences in the North and South: In the North, forest grazing intensity declined from 1–14% in 1930 to almost 0% until 2007, while in the South it remained highest of all regions throughout the period between 2–6% (1910) and 1–13% (2000). Overall, national as well as regional results suggest a general tendency of livestock holders to abandon open grazing on pastures and particularly in forests, made visible by the substitution of directly grazed biomass with hay and feed

crops.

3.2. ES demand for fuelwood use

Our results show how fuelwood extraction declined substantially while final energy consumption increased in parallel, due to profound shifts in the types of resources used. Total final energy consumption (TFEC; fuelwood, coal, oil, natural gas, and electricity) in the U.S. grew 17-fold from 1870–2012 (Fig. 5a). In 1870, forests provided for more than 80% of the total TFEC. Until 1900, fuelwood remained the major source of technical energy in the country (Clawson, 1979; Schurr, 1984; Williams, 1987) but it was increasingly replaced by coal already by 1910. Forest biomass contributed between 1.5 and around 3 exajoules (EJ/yr) to TFEC during the 142 years studied. In 2012 fuelwood accounted for 4% of the TFEC in the U.S. (Fig. 5a). The large increase in TFEC, especially after 1940, was driven by unprecedented levels of fossil fuel consumption (mainly oil and natural gas) but also of final electricity, which in the U.S. is mostly generated from burning coal (International Energy Agency, 2019).

In the 19th century fuelwood was used in the transport sector to some extent to move steamboats and locomotives but also for industrial purposes (mainly heating) (Warde, 2019). The residential sector was by far the main consumer of fuelwood in the U.S. for the first 90 years of the observed period (Birdsey et al., 2006; MacCleery, 1993; Morrison, 1992) but was overtaken by the industrial sector after the 1960s (Fig. 5b).

Disaggregating TFEC into the major end-use sectors (Fig. 6a–d) reveals profound changes in the importance of forest provisioning ES for societal energy use. Large-scale substitution of fuelwood with fossil fuels took place in all sectors, but on different levels. Fuelwood was important in the residential sector until 1930 (Fig. 6a) but declined significantly from 1940–1970, while natural gas and oil consumption increased. Between 1970 and 1980, fuelwood consumption in households more than doubled, from 423 Petajoule per year (PJ/yr) to almost 900 PJ/yr, a direct response to the oil price shocks of the 1970s (Song et al., 2012). During the Great Depression period in the 1930s, and from 2007 to 2012 (financial crises), fuelwood consumption increased slightly as well. The commercial and public services sector (Fig. 6b) displays a similar energy mix as the residential sector but fuelwood was practically irrelevant already after 1910.

In the transport sector (Fig. 6c) in 1870, fuelwood accounted for almost 60% of TFEC, while a decade later this share had shrunk to below 30% and declined to almost zero by 1910, since fuelwood used in locomotives and steamboats had almost completely been replaced by coal. For the rest of the observed period, oil products (mostly gasoline for vehicles) provided for 70–97% of total final energy in this sector. In the industrial sector (Fig. 6d), use of fuelwood declined from almost 40% of TFEC in 1870 to 6% in 1930. However, fuelwood use in this sector never disappeared and even increased after 1950, accounting for 10% of TFEC in 2012. In the first third of the observed period, industrial energy uses of wood occurred mainly in charcoal making for the iron & steel industry. In the following decades, the paper industry and later the wood manufacturing industries emerged as main consumers, where fuelwood was mainly used for heating, and later for self-sufficient electricity generation (Fernandes et al., 2007; Schulz, 1993; Skog & Rosen, 1997).

4. Discussion

4.1. Understanding the drivers of the U.S. forest transition

Our study presents long-term changes in ES capacity, provisioning flows and demand of U.S. forests. Results show that the change of ES capacity was enabled through shifts in ES demand, accompanied by externalisation of provisioning ES. We estimated that total forest HANPP_{harv} accounted for around 20% of total forest NPP_{pot} in 1890 (Fig. 2b). This indicates high harvest pressure on forests in the U.S. in the late 19th and early 20th century, close to the thermodynamically possible

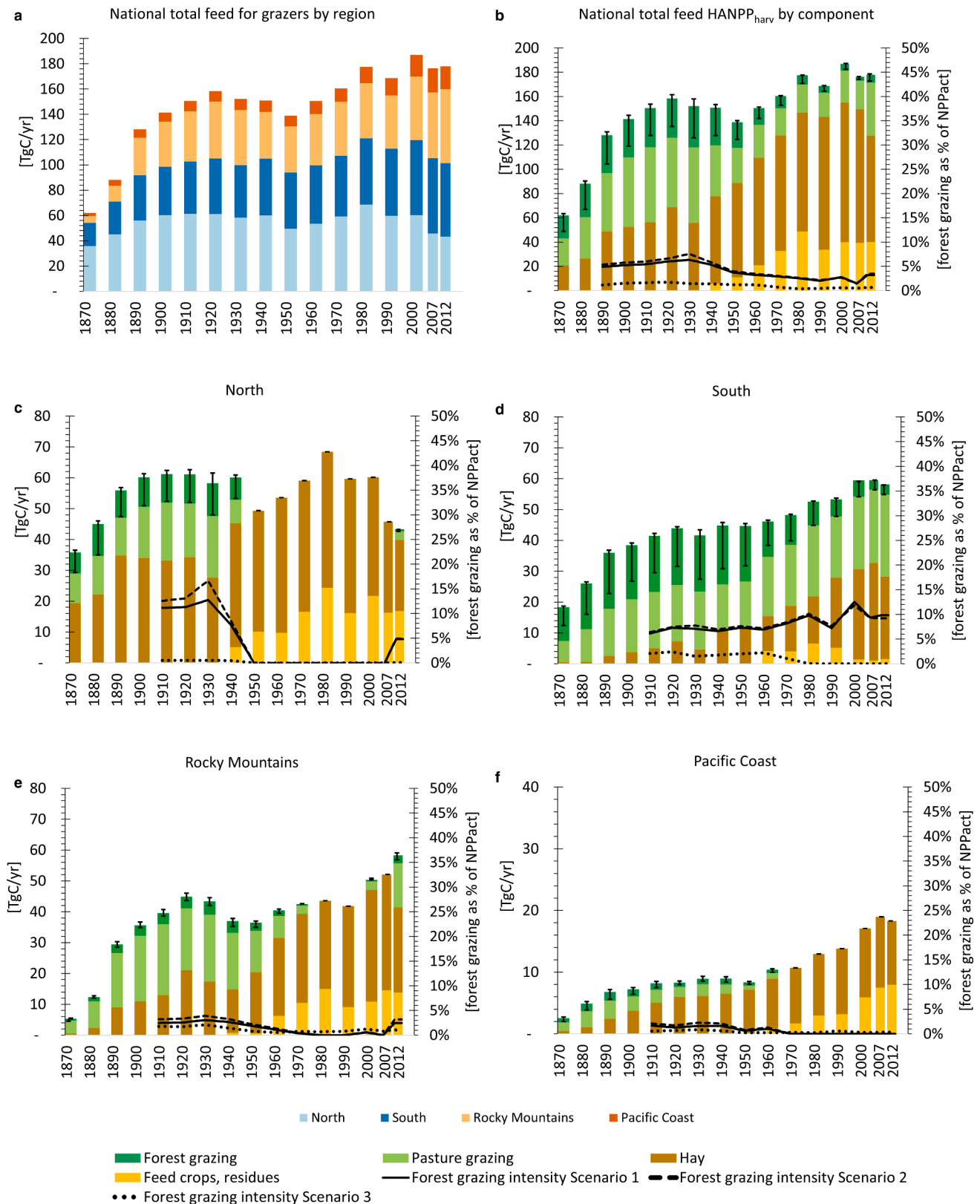


Fig. 4. Total feed for grazers: National total **a** by sub-regions **b** by components and grazing intensity; **c-f** Regional feed for grazers by components and forest grazing intensities; **c** North **d** South **e** Rocky Mountains **f** Pacific Coast; Feed components for **b-f**: biomass grazed in forests and pastures, hay, feed crops including residues [primary axis, columns, TgC/yr]; Forest grazing intensity for Scenarios 1-3 as % of total NPP in forests [secondary axis, lines, %]; Notes: Forest NPP values available from 1890 (national scale, **b**), and 1910 (regional scale, **c-f**); Error bars (**b-f**) indicate minimum and maximum variations for forest grazing caused by the three scenario estimates; Note the different primary y axis in **f**. Conterminous U.S. 1870-2012.

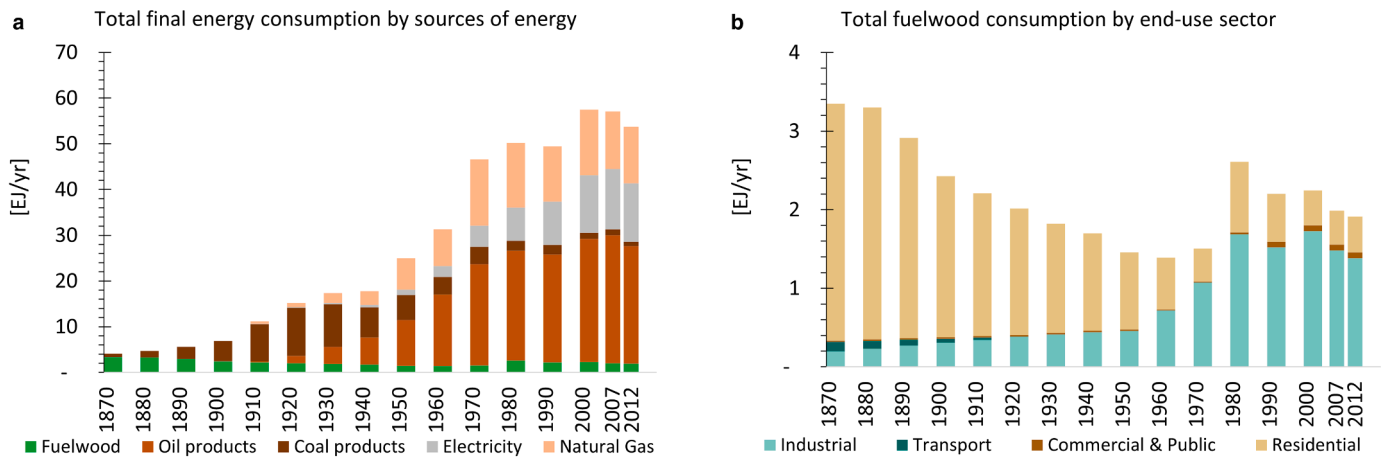


Fig. 5. Changes in final U.S. energy consumption **a** Total final energy consumption by sources of energy (coal, oil, natural gas, fuelwood, electricity); **b** Total fuelwood consumption by end-use sector, absolute numbers. Conterminous U.S. 1870-2012

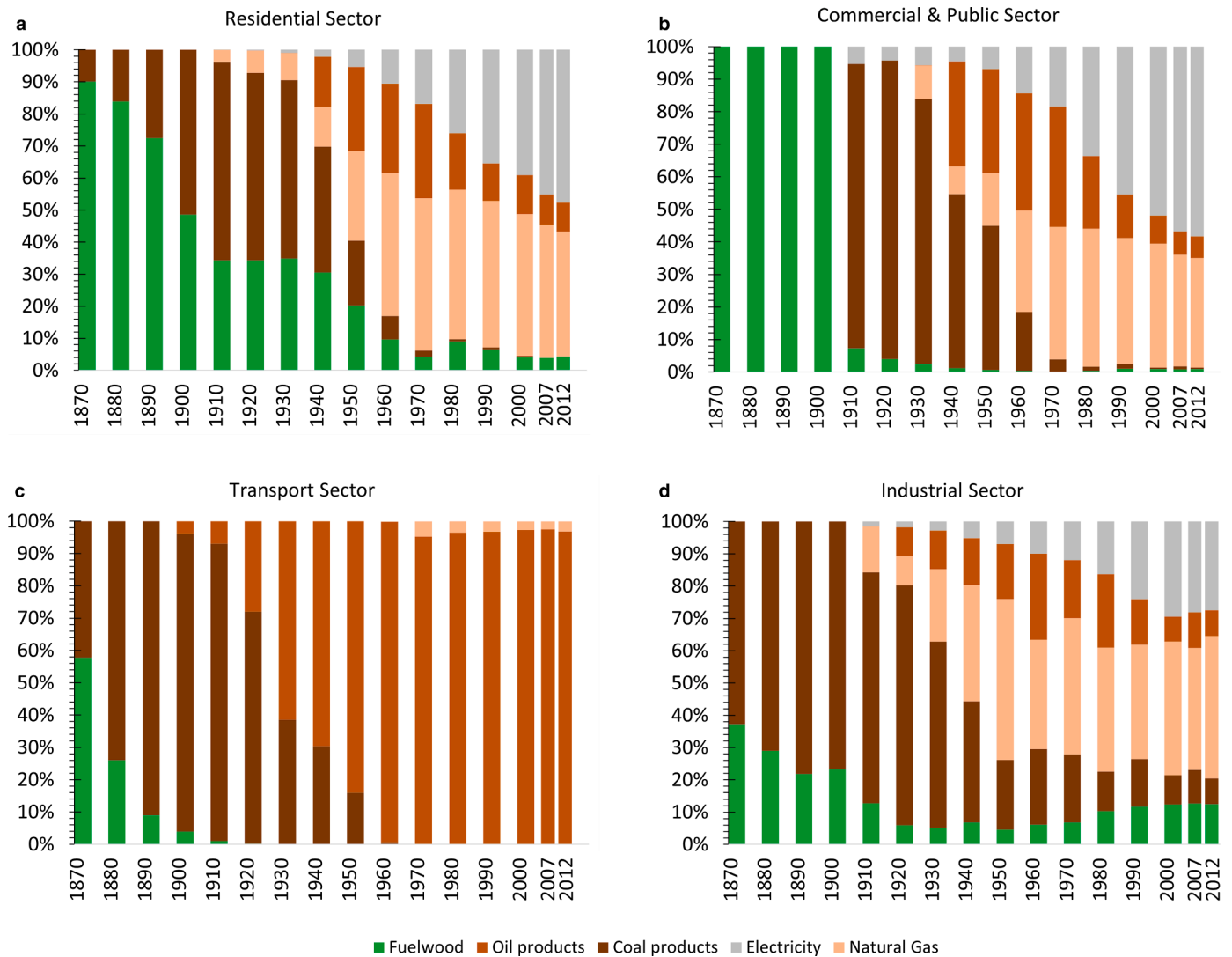


Fig. 6. Sector-wise shares of sources of energy in total final energy consumption per year. **a**) Residential **b**) Commercial **c**) Transport **d**) Industrial, Conterminous United States, 1870-2012

maximum aboveground wood harvest of 30% of NPP_{pot} estimated by Schulze et al. (2012). Even though total forest $HANPP_{harv}$ did not continuously decline but rather fluctuated in the studied period, it eventually decreased to around 13% of NPP_{pot} in 2012. In addition, despite pronounced growth in the number of grazers and total final energy consumption, demand for the respective resource-intensive forest provisioning ES did not expand because they were increasingly substituted by other forms of material and energy (mainly fossil fuels, and biomass, produced in a fossil-powered agricultural system). These observed dynamics in ES supply and demand enabled biomass C-stocks in forests to almost double between 1910 and 2012, reflecting an increase in ES capacity (Fig. 2a). Our results thus highlight the tight relationship between supply of, demand for, and capacity of forest ES: shifts in the demand for provisioning ES from mixed to industrial wood use, accompanied by spontaneous reforestation, intensive timber management, tree plantations, and enhanced biomass growth rates due to environmental change fostered forest biomass growth. Hence, in combination with the large-scale resource substitution processes described in this study, the overall long-term reduction in specific provisioning ES added to the increase of forest ES capacity i.e., biomass C stocks (Fig. 2a).

4.2. Externalization of demand for forest grazing

We presented a novel approach for approximating the impact of grazing on forest NPP on the national and sub-national scales. However, determining the exact impact of forest grazing is difficult, as it depends on many site-specific factors such as stocking density, climate, elevation, species and age of trees, forest management etc. (Öllerer et al., 2019). While there is a large body of literature analysing the local impacts of forest grazing on biomass, results vary substantially and present to some extent conflicting implications Covington et al., (1994; Öllerer et al., (2019). Some studies found positive effects of forest grazing for e.g., biomass growth, species composition, and biodiversity (Galleguillos et al., 2018; McEvoy et al., 2006; Shakeri et al., 2012), whereas others reported the opposite, and other negative effects such as soil erosion or -compaction (Eldridge et al., 2016; Krzic et al., 2006; Ratovonamana et al., 2013; Sharrow, 2007). We could not assess such specific negative feedbacks on the ES capacity of forests but instead showed how forest grazing intensities may considerably diverge from national trajectories on sub-national scales. In three of four regions, the availability of large amounts of hay and feed crops basically made forest grazing obsolete after WWII. This is both a direct consequence of the pronounced shrinking of total grazed forest areas, caused by increasing rates of regulations and restrictions during the 20th century (Potter, 1913; Steen, 2004) and industrialisation of the livestock sector: Its biophysical basis in the U.S. changed dramatically over the last 100 years, as geographically concentrated intensive high protein feed crop-based operations (e.g., feedlots) have largely replaced extensive (forest) grazing (Bouwman et al., 2005; Hart and Mayda, 1998; Herrero et al., 2013; Milbrandt et al., 2018). Forest grazing intensity remained between 1 and 13% of NPP_{act} only in the Southern states. The overall decline of forest grazing, especially in the Eastern U.S., can be associated with the westward expansion, as agriculture in the Mid-West emerged as more profitable and efficient than in the East. On the one hand, this fostered reforestation in the South, as large timber companies established intensively managed commercial timber plantations, but on the other hand displaced smallholder livestock farming (Foster et al., 1998; Gregg, 2010; Grelen, 1978).

4.3. Externalization of demand for fuelwood

Fuelwood extraction on the national scale decreased substantially throughout the observed period. Including substitution processes, we consistently quantified shifts in the ES demand for energy in different socioeconomic sectors, related to industrialisation processes in parallel

to the forest transition. We showed that fuelwood was central for meeting the final energy demand of the U.S. in the late 19th century but was replaced by fossil fuels and made practically irrelevant after 1910 (Fig. 5a). The ES demand not only moved from the residential to the industrial sector, but also from fuelwood to external substitutes, first to coal and then to oil and natural gas (Figs. 5 and 6).

4.4. Effects of externalising forest ES

The observed externalisation of provisioning ES demand from forests to agricultural land and fossil energy sectors in the U.S. had important implications for the use of forest biomass, which manifested in increasing harvest of industrial wood, while still enabling a significant recovery of forest biomass C stocks. However, these developments were only possible because of large-scale substitution processes: Intensification of agricultural feed crops and livestock production systems, enabled through the massive use of fossil fuels for artificial fertiliser, mechanisation, improved modes of transportation (e.g., establishment of railroads), and foreign trade enabled significant growth of output without expansion of agricultural land, making side-uses of forests such as livestock grazing practically irrelevant. The large-scale substitution of fuelwood with coal, oil, and gas led to unprecedented levels of final energy consumption. These processes can be related to the fundamental transformation from an agricultural to a fossil-fuel powered industrial society during the 20th century in the U.S. (Gierlinger and Krausmann, 2012; Steinhart and Steinhart, 1974). Although analysing the broader historical societal rationales that strove the evolution of socio-ecological variables investigated here was beyond the scope of this study, we highlight three major alternative lines of thought possibly explaining the societal transformation of shifts in ES demand and the substitution of fuelwood with fossil fuels: 1) A market-driven explanation: Scarcity of wood drove the shift from fuelwood to more abundant fossil fuels (Barnett and Morse, 2013; Mather, 2001). 2) A profit-driven explanation: due to their higher energy efficiency, profitability of fossil fuels exceeded by far that of fuelwood, thus urging its replacement (Atkinson and Halvorsen, 1976; Huppmann and Egging, 2014). 3) A power-driven explanation: the physical nature of fossil fuels made them more suitable for the need to control and discipline a large workforce under a new phase of industrial capitalism (Malm, 2016). Additionally, the completion of the first transcontinental railroad in the U.S. in the 1870s enabled the establishment of a national market, requiring the mobility of the labour force, raw materials for the industry, and commodities (Hobsbawm, 2010).

Changes in patterns of production and consumption related with these processes were accompanied by a surge in associated greenhouse gas emissions like CO_2 , CH_4 , N_2O (Environmental Sciences Division, 2010; Erb et al., 2007; Gingrich et al., 2019). Although U.S. forests sequester substantial amounts of carbon, the country is still one of the largest emitters of direct and indirect carbon (He and Hertwich, 2019): From 2007 to 2012, C emissions caused by fossil fuel use alone accounted for a flux of around 1500 TgC/yr (Friedlingstein et al., 2020) while forests sequestered 716 TgC/yr during the same period (Magerl et al., 2019). Although forest thickening contributes to climate change mitigation and biodiversity conservation, the societal transformation enabling these developments resulted in the emergence of a largely unsustainable socioeconomic system relying on unprecedented levels of energy and material throughput, which might be counteracting the positive effects of the forest transition. In this context, it has been repeatedly argued that the increased use of biofuels (renewable energy) and harvested wood products (long-term C storage) is an appropriate strategy for climate change mitigation. In spite of the undeniable potential of these measures to counteract some C emissions, several studies have highlighted the possible unintended negative side-effects such approaches could have: Harvesting more wood in order to transfer C from forests to products with shorter lifespans than old-growth forests, and increased biofuel use, could spur rejuvenation of forests, decreasing

their C residence times and reducing the storage function of forest biomass (e.g., Braun et al., 2016; Erb et al., 2007; Harmon, 2019; Hudiburg et al., 2019). Hence, we highlight the importance to consider and systematically analyse substitution processes and shifts in resource use associated with long-term forest change.

4.5. Limitations and uncertainty discussion

The HANPP framework offers a consistent approach to investigate long-term land-use change based mostly on empirical statistical data. The consistent integration of different processes of biomass extraction from forests in the HANPP framework enables to quantify the relative importance of different ES but does not assess the specific feedbacks exerted by these processes. Our empirical focus on the national-level, and some regional trends, linking HANPP to changes in resource use, does not capture detailed local-scale dynamics of land-use and forest change (e.g., Jeon et al., 2014; Niedertscheider et al., 2017). We compared historical U.S. forestry statistics used here with previously published studies (Fernandes et al., 2007; Gierlinger and Krausmann, 2012) and found no significant differences between sources (section S5). Thus, we assume that the statistical records used here are reasonably accurate. We conducted a sensitivity analysis (section S7) to acknowledge possible bias and variations in historical data (Krausmann et al., 2013): This analysis revealed that modifying the main input factors resulted in maximum deviations in total forest $\text{HANPP}_{\text{harv}}$ of -23% to +50% in 1870 and -12 to +30% in 2012, possibly attributable to statistical differences in historical data and underlying methodological assumptions. Deviation in cumulative wood harvest (fuelwood and industrial wood) where found to be the cause of the largest uncertainty, resulting in deviations of -19 and +49% in 1870 and -11 to +30% in 2012. By contrast, testing the input factors for the forest grazing estimation alone revealed the robustness of these results (maximum deviation of -6% to +3% in 1870 and -1% to +0.5% in 2012). The range of variation caused by assuming no increase in NPP_{pot} was found not to be significantly influencing our main result for total forest $\text{HANPP}_{\text{harv}}$ as percentage of total forest NPP_{pot} (maximum deviation of -3.6% in 1900 under constant 1990 NPP_{pot} , and +3.9% in 1990 under constant 1890 NPP_{pot} assumption). Overall, the sensitivity analysis showed that our study may even have underestimated the impact of demand for ES provision on forest NPP. Despite the possible deviations inflicted by using historical sources and the underlying methodological assumptions, the sensitivity analysis revealed no significant changes in the temporal trends described in this study and thus validates the robustness of our results against the uncertainty in the coefficients applied.

5. Conclusions

This study quantified ES provided by forests in the U.S., identifying links between changes in societal resource use, C fluxes, and the biophysical functions of forests for society, an aspect rarely addressed in forest transition research. By applying a long-term perspective, we showed that substitution of forest provisioning ES played a central role in the U.S. FT, emphasising the need for analyses of the temporal trajectories of ES in connection to changes in social metabolism, substitution processes and environmental change. Our results show that changes in provisioning ES offer a possible explanation for the apparent contradiction of both increasing levels of wood harvest and forest biomass stocks. Changes in demand for provisioning services were enabled by far-reaching shifts in resource and land use: fuelwood was replaced by fossil fuels, while forest biomass for grazing was substituted by agricultural crops and feed from grasslands. Thus, supply of ES was increasingly externalised from forests. Among other factors, these shifts in ES supply enabled forest biomass stock recovery, and therefore led to an increase of the forest C sink. Due to the focus of this study, we did not explicitly assess other possible explanations for forest recovery such as increased NPP under altered growth conditions, including fertilization

by nitrogen deposition, longer growing season and CO_2 enrichment (Friedlingstein et al., 2020; Hickler et al., 2008; Hubau et al., 2020; Tagesson et al., 2020; Zhu et al., 2016). Systematically integrating these processes into future studies will be central to a better understanding of the interrelations between social metabolism, natural change, and ecosystem dynamics.

In conclusion, we argue that a socio-ecological perspective on forest use enables re-embedding the social drivers of the forest transition within their biophysical environment, thus contributing to a better understanding of the links between forest ecosystems and processes of resource use influencing respective C dynamics. Our findings have strong implications for the design of future climate change mitigation options (e.g., re- and afforestation, agroforestry, substitution of fossil with biofuels) as it stresses the needs to develop systemic models addressing the interlinkages between provisioning (e.g., energy and food provisioning), and regulating (e.g., C sequestration) ecosystem services, and between forest- and resource use in general: We argue that policymakers need to acknowledge the obvious contradictions and challenges arising for the possible transition to a sustainable bioeconomy. Among other measures, incentivising forest regrowth to increase the terrestrial C sink, enhancing resource use efficiency, and substituting fossil fuels with renewable energy will be important steps towards climate change mitigation. At the same time, the proposed increased use of (forest) bioenergy should not be considered as a sustainable energy alternative. Supplying the current societal resource throughput to sustain industrial lifestyles will hardly be possible by substituting fossil fuels with biofuels. Instead, strategies need to be developed that reconcile increasing C-sinks and lowering GHG emissions to meet international climate change mitigation agreements with a shift away from resource-intensive consumption patterns.

CRedit authorship contribution statement

Andreas Magerl: Conceptualization, Methodology, Validation, Formal analysis, Data curation, Writing – original draft, Visualization. **Sarah Matej:** Data curation, Writing – review & editing. **Lisa Kaufmann:** Data curation, Writing – review & editing. **Julia Le Noë:** Data curation, Writing – review & editing. **Karlheinz Erb:** Data curation, Writing – review & editing. **Simone Gingrich:** Conceptualization, Methodology, Formal analysis, Writing – original draft, Writing – review & editing, Supervision, Project administration, Funding acquisition.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Supplementary materials

Supplementary material associated with this article can be found, in the online version, at [doi:10.1016/j.resconrec.2021.105927](https://doi.org/10.1016/j.resconrec.2021.105927).

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